

2.4 Boolean logic

How to run this unit, lesson by lesson

Contents

[Start here](#)

[Preparation and what you need](#)

[Using the three resources together](#)

[Evidence, checking and common difficulties](#)

[The lesson plans, in teaching order](#)

[Teacher answers](#)

Start here

5 lessons. A realistic estimate is **5 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Logic gate lab (b1-l01)	1	Lesson_Plans/BL_L01_LessonPlan.pdf
2	Truth table detectives (b1-l02)	1	Lesson_Plans/BL_L02_LessonPlan.pdf
3	Circuit workshop (b1-l03)	1	Lesson_Plans/BL_L03_LessonPlan.pdf
4	End-of-topic test (b1-l04-test)	1	Lesson_Plans/BL_L04_TEST_LessonPlan.pdf
5	Logic sprint (b1-bonus)	1	Lesson_Plans/BL_BONUS_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (teacher.html), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at topics.html with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- no optional challenge activity – b1-101, b1-102, b1-103

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Logic gate lab b1-101

How do you draw a logic diagram?

OUTCOMES	know the symbols for the logic gates AND, OR and NOT; understand the truth tables for AND, OR and NOT; combine Boolean operators using AND, OR and NOT
STATIONS AND ACTIVITIES	6 stations, 15 activities, 24 marks (6 guided, 9 independent, 0 challenge)
RESOURCES	BL_L01_LogicGateLab_Lesson.pptx, BL_L01_LogicGateLab_Worksheet.docx
PLAN	Lesson_Plans/BL_L01_LessonPlan.pdf

2. Truth table detectives b1-102

How do you complete a truth table?

OUTCOMES	understand how to complete truth tables from one- and two-level logic diagrams
STATIONS AND ACTIVITIES	6 stations, 11 activities, 38 marks (5 guided, 6 independent, 0 challenge)
RESOURCES	BL_L02_TruthTableDetectives_Lesson.pptx, BL_L02_TruthTableDetectives_Worksheet.docx
PLAN	Lesson_Plans/BL_L02_LessonPlan.pdf

3. Circuit workshop b1-103

How do you create logic diagrams from truth tables?

OUTCOMES	understand how to create, complete or edit logic diagrams and truth tables for given scenarios
STATIONS AND ACTIVITIES	6 stations, 14 activities, 35 marks (5 guided, 9 independent, 0 challenge)
RESOURCES	BL_L03_CircuitWorkshop_Lesson.pptx, BL_L03_CircuitWorkshop_Worksheet.docx
PLAN	Lesson_Plans/BL_L03_LessonPlan.pdf

4. End-of-topic test b1-104-test

Your 2.4 test is done on paper. Warm up with Logic sprint first, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	BL_L04_EndOfTopicTest_Reflection_Worksheet.docx
PLAN	Lesson_Plans/BL_L04_TEST_LessonPlan.pdf

5. Logic sprint b1-bonus

Two minutes, random circuits and expressions, speed bonuses and streaks. Replay as often as you like to beat your personal best.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.
STATIONS AND ACTIVITIES	0 stations, 1 activities, 0 marks (0 guided, 1 independent, 0 challenge)
RESOURCES	none recorded
PLAN	Lesson_Plans/BL_BONUS_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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